

SUMMARY

I am a software engineer with 10+ years shipping distributed systems for manufacturing, financial services, and real-time operations — including MES at MachineMetrics and platforms deployed in offshore oil and gas. I specialize in TypeScript/JavaScript full-stack engineering, always-on shop-floor UIs, and AI systems (LLM orchestration, VLM pipelines, tenant isolation, OWASP LLM Top 10 guardrails), delivered through spec-driven development with agent workflows. Clients rely on me for pragmatic architecture, measurable performance on data-heavy interfaces, and clear collaboration with distributed international teams.

MOST AMAZING PLATFORM

Hardened MachineMetrics’ React MES — a 24/7, data-heavy manufacturing execution product on high-volume IoT + ERP streams — shipping Waiting For Work and palletized-machine workflows where a stalled UI stops the floor, not just a tab.

EXPERIENCE

MachineMetrics — Full Stack Software Engineer

Apr 2024 – Present

- Delivered React MES for CNC operators on a high-volume IoT + ERP platform — features that map how shops schedule, execute, and recover work mid-shift.
- Shipped Waiting For Work, palletized machines, FAI, labor reporting, part counting, scheduling, and ETTC across ERP-driven, labor-based, and legacy production models.
- Hardened 24/7 shop-floor UIs with WebSockets, web workers, Redux-style state, and microservice middleware isolating telemetry ingest from domain rules and presentation.
- Kept live grids usable under bursty machine events via list virtualization and memoized selective re-renders — avoiding full-tree refreshes on every update.
- Owned APIs, migrations, and production incidents end to end; drove SDD with agent orchestration so product intent stayed explicit under manufacturing constraints.
- Bridged ERP intent and machine reality so idle time, pallet queues, first-article checks, and labor stayed coherent for operators mid-shift.

React · Node.js · TypeScript · WebSockets · Workers · MySQL · Redis · NATS · Hasura · GraphQL

Creed Interactive · Mariner — Software Engineer / Consultant

Jul 2021 – Apr 2024

- At Mariner, optimized always-on Angular shop-floor apps so continuous industrial telemetry did not freeze operator screens during peak load.
- Cut render storms with CDK-style virtual scrolling, OnPush change detection, WebSockets, workers, and NgRx — middleware kept ingest pipelines out of the UI process.
- At Creed, led Keycloak SSO migration and a super-app navigation layer so multiple Angular/React MFEs shared one identity boundary without breaking product switching.
- Delivered NestJS services and Dockerized platform pieces with inventory, logistics, and operations stakeholders who needed stable day-to-day industrial tooling.
- Treated shop-floor latency as a production risk — jank reads as lost floor trust; SSO and MFE tenancy had to agree for suites to feel unified.

Angular · React · NgRx · WebSockets · Workers · NestJS · Keycloak · Docker · Azure

B2ML Sistemas — Full Stack Software Engineer / Technical Lead

Jan 2019 – Jul 2021

- Led a banking microservices platform that linked dozens of transit recharge providers while satisfying financial compliance and audit expectations.
- Homologated a POS from scratch and integrated banking/payment partners under COVID deadline pressure without relaxing security gates.
- Mentored engineers and turned customer payment workflows into shippable C#/.NET and Angular releases with clear ownership boundaries.
- Treated provider outages, POS homologation, and banking rules as the product itself — not side chores around the API.

C# · .NET Core · RabbitMQ · PostgreSQL · Angular · AWS · Docker

RZX Tecnologia — Full Stack Software Engineer / Technical Lead

Jan 2016 – Dec 2018

- Owned the full industrial product — Angular/React operator UIs and backend services — for real-time offshore operational data from domain model through deploy.
- Designed 24/7 surfaces with WebSockets, workers, shared state, and microservice middleware so ingest spikes could not stall the control UI.
- Kept live operational grids usable with list virtualization and OnPush / memoized rendering under continuous field updates.
- Shipped a platform selected for offshore oil and gas; later expanded as Cognitiv and still runs — continued as consultant after the transition.

Node.js · Angular · React · WebSockets · Workers · Docker · Redis · PostgreSQL

SaskPower — Software Developer Intern

May 2015 – Aug 2015

- Built ISO-compliant hybrid mobile apps for internal utility operations during a summer internship in Regina, Saskatchewan, Canada.
- Delivered Cordova and JavaScript clients used by field and office staff for day-to-day utility workflows under ISO process constraints.
- Collaborated with Canadian utility stakeholders to ship usable tools in one summer term without sacrificing process compliance.

Apache Cordova · jQuery · JavaScript · HTML/CSS

PUBLIC DEMOS

Promptdesk
app.promptdesk.ferredemo.dev
github.com/Guilheeeeeerme/promptdesk

Argus
app.argus.ferredemo.dev
github.com/Guilheeeeeerme/argus

Quizzeira
app.quizzeira.ferredemo.dev
github.com/Guilheeeeeerme/quizzeira

KEY ACHIEVEMENTS

MachineMetrics MES — critical shop-floor product
Always-on MES on IoT + ERP: WFW, palletized machines, FAI, labor, scheduling — live UIs that cannot stall mid-shift under bursty machine data.

Offshore oil & gas platform (RZX – Cognitiv)
Built the core industrial platform end to end; selected for offshore deployment, later expanded as Cognitiv, still running in production.

Fintech payments lead (B2ML)
Led microservices transit payments, POS homologation from scratch, and banking security under COVID-driven deadline pressure.

TECHNICAL SKILLS

FEATURED
TypeScript, React, Angular, Node.js, NestJS, LLM Systems, Spec-Driven Development, Claude, Cursor, OpenCode, Subagents, Microservices

LANGUAGES
JavaScript, TypeScript, Java, Python, C#/.NET

FRONTEND
React, Angular, Redux, NgRx, Micro Frontends, Virtualization, OnPush / memoized renders

BACKEND
Node.js, NestJS, Express, Fastify, FastAPI, .NET Core, REST, GraphQL, Hasura, Keycloak

DATA & MESSAGING
MySQL, PostgreSQL (RLS), Redis, RabbitMQ, NATS, BullMQ, WebSockets

INFRA
Docker, Git, Kubernetes (basic), AWS, Azure, GitHub Actions, MinIO, Workers

ARCHITECTURE
DDD, Event-Driven Architecture, Microservices, Distributed Systems, MES

AI SYSTEMS
LLM orchestration, multi-provider failover, VLM pipelines, rate limiting, tenant isolation, OWASP LLM Top 10, token/cost controls

AGENT DELIVERY
SDD, PRD, Skills, Worktrees, Parallel Development, AI PR Review

EDUCATION

MBA in Software Engineering with AI
Faculdade Full Cycle de Tecnologia (FCTECH), Brazil
2026 – present

Master's Candidate — Computer Science & Technology (incomplete)
Federal University of Itajubá (UNIFEI), Brazil
2019 – 2021

Bachelor of Information Systems
Federal University of Itajubá (UNIFEI), Brazil
2011 – 2016

LANGUAGES

English — Fluent
Portuguese — Native